

P₁₋₂ 习题8 P₄

$$I) f(x, y) = t^2 x^2 + t^2 y^2 - t^2 xy \tan \frac{\pi}{4}$$

$$II) f(x+y, x-y, xy) = (x+y)^{xy} + xy^{x^2}$$

$$III) \begin{cases} X = x+y \\ Y = \frac{y}{x} \end{cases} \rightarrow \begin{cases} x = \frac{X}{1+Y} \\ y = \frac{XY}{1+Y} \end{cases}$$

$$f(X, Y) = x^2 y^2 = \frac{X^2(1-Y^2)}{(1+Y)^2}$$

$$f(x, y) = \frac{x^2(1-y^2)}{(1+y)^2} = \frac{x^2(1-y)}{1+y}$$

$$IV) z(x, y) = 1 + f(\sqrt{x} - 1) = x$$

$$f(\sqrt{x} - 1) = x - 1$$

$$\text{令 } X = \sqrt{x} - 1 \rightarrow x = (X+1)^2$$

$$f(X) = (X+1)^2 - 1 \rightarrow f(x) = x^2 + 2x$$

$$z(x, y) = \sqrt{y} + x - 1$$

P₃ 习题8 B

因为 $x \neq 0$, 取 $t = x^{-1}$

$$f(tx, ty) = f(x^{-1} \cdot x + x^{-1} y) = x^{-k} f(x, y)$$

$$\text{又有: } f(x^{-1} \cdot x + x^{-1} y) = f(1, \frac{y}{x})$$

$$x^{-k} f(1, \frac{y}{x}) = f(x, y) = z$$

$$\text{令 } F(\frac{y}{x}) = f(1, \frac{y}{x}) \rightarrow z = x^k F(\frac{y}{x}) \quad \square$$

P₄ 题8 P₆ 1, 2

$$I). \lim_{(x,y) \rightarrow (0,0)} \frac{2 - \sqrt{x+y+4}}{x+y} = \lim_{(x,y) \rightarrow (0,0)} \frac{-(x+y)}{(x+y)(2 + \sqrt{x+y+4})} = \lim_{(x,y) \rightarrow (0,0)} \frac{-1}{2 + \sqrt{x+y+4}} = -\frac{1}{4}$$

$$II) \lim_{(x,y) \rightarrow (0,0)} \frac{(2+x)(1+y)}{xy} = \lim_{(x,y) \rightarrow (0,0)} \frac{(2+x)y}{xy} = \lim_{(x,y) \rightarrow (0,0)} (2+x) = 2$$

P₅ 题8 P₇

$$I) f(x,y) = \begin{cases} x \sin \frac{1}{y} + y \sin \frac{1}{x} & , xy \neq 0 \\ 0 & , xy = 0 \end{cases}$$

$$\lim_{(x,y) \rightarrow (0,0)} x \sin \frac{1}{y} + y \sin \frac{1}{x} = \lim_{(x,y) \rightarrow (0,0)} x \sin \frac{1}{y} + \lim_{(x,y) \rightarrow (0,0)} y \sin \frac{1}{x}$$

由于 $\sin \frac{1}{y}$, $\sin \frac{1}{x}$ 是有界量, x, y 是 $(x,y) \rightarrow (0,0)$ 时的无穷小量

$$\text{故: } \lim_{(x,y) \rightarrow (0,0)} x \sin \frac{1}{y} = 0, \quad \lim_{(x,y) \rightarrow (0,0)} y \sin \frac{1}{x} = 0$$

$$\text{即 } \lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$$

$$II). f(x,y) = \begin{cases} \frac{xy}{x+y} & , xy \neq 0 \\ 0 & , xy = 0 \end{cases}$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x+y}$$

$$\text{令 } y = x \quad \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x+y} = \lim_{x \rightarrow 0} \frac{x^2}{2x} = 0$$

$$\text{令 } y = x(x-1) \quad \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x+y} = \lim_{x \rightarrow 0} \frac{x^2(x-1)}{x^2} = \lim_{x \rightarrow 0} x-1 = -1$$

不同路径极限不等 $\rightarrow$ 极限不存在.

P₆₋₇ 题8 P₈

$$I) z = \frac{1}{\sqrt{x^2+y^2}}$$

在 $(0,0)$ 点, $z \rightarrow +\infty$, 函数间断

$$II) z = \frac{1}{\sin \pi x} + \frac{1}{\sin \pi y}$$

在直线 $x=a, y=b$ ($a, b \in \mathbb{Z}$) 处, 函数间断

$$III) z = \frac{2x+y^2}{2x-y^2}$$

在 $y^2=2x$ 抛物线上, 函数间断.

$$IV) u = \frac{1}{xy z}$$

$xyz=0$ 时 ($x=0$ 或 $y=0$ 或 $z=0$) 时, 函数间断.

P₈ 习题 8 P₉ 4

$$f(x, y) = x(y-1) \arcsin \sqrt{\frac{y}{y-1}}$$

$$y=1 \text{ 时}, f(x, 1) = x, f_x(x, 1) = 1$$

P₉₋₁₀ 习题 8 P₁₀ 4, b, 8

$$IV) z = \frac{e^{\pi y}}{e^x + e^y}$$

$$\frac{\partial z}{\partial x} = \frac{ye^{\pi y}(e^x + e^y) - e^{\pi y + x}}{(e^x + e^y)^2} = \frac{e^{\pi y}(ye^x - e^x + ye^y)}{(e^x + e^y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{\pi e^{\pi y}(e^x + e^y) - e^{\pi y + y}}{(e^x + e^y)^2} = \frac{e^{\pi y}(\pi e^x + \pi e^y - e^y)}{(e^x + e^y)^2}$$

$$VI) z = \arcsin(3-2\pi y) + \sin(3-\frac{2x}{y})$$

$$\frac{\partial z}{\partial x} = \frac{-2y}{\sqrt{1-(3-2\pi y)^2}} - \frac{2}{y^2} \cos(3-\frac{2x}{y})$$

$$\frac{\partial z}{\partial y} = \frac{-2x}{\sqrt{1-(3-2\pi y)^2}} + \frac{2x}{y^2} \cos(3-\frac{2x}{y})$$

$$VII) z = (1+\pi y)^{\pi y}$$

$$\ln z = (\pi y) \ln(\pi y + 1)$$

$$\text{两边对 } x \text{ 求偏导: } \frac{\partial z}{\partial x} = \ln(\pi y + 1) + (\pi y) \frac{y}{1+\pi y}$$

$$\frac{\partial z}{\partial x} = (1+\pi y)^{\pi y} \left[\ln(\pi y + 1) + \frac{y(\pi y)}{1+\pi y} \right]$$

同理, 两边对 y 求偏导: $\frac{\partial z}{\partial y} = \ln(xy+1) + (x+y) \cdot \frac{1}{1+xy}$

$$\frac{\partial z}{\partial y} = (1+xy)^{xy} \left[\ln(xy+1) + \frac{x(1+xy)}{1+xy} \right]$$

P₁₁₋₁₂ 习题8 P₁

I). $C: \begin{cases} z = \frac{x^2+y^2}{4} \\ y=4 \end{cases} \quad k = z_x(2,4) = \frac{x}{2} = 1$

$$\alpha = \arctan k = \frac{\pi}{4}$$

II) $C: \begin{cases} z = \sqrt{1+x^2+y^2} \\ x=1 \end{cases} \quad k = z_y(1,1) = \frac{y}{\sqrt{1+x^2+y^2}} \Big|_{(1,1)} = \frac{1}{\sqrt{3}}$

$$\vec{s} = (0, \sqrt{3}, 1)$$

$$C: \frac{x-1}{0} = \frac{y-1}{\sqrt{3}} = \frac{z-1}{1}$$

$$\pi: \sqrt{3}y + z - 2\sqrt{3} = 0$$

III) $C_1: \begin{cases} z = x + \frac{y^2}{6} \\ y=2 \end{cases} \quad C_2: \begin{cases} z = \frac{x^2+y^2}{3} \\ y=2 \end{cases}$

联立 C_1, C_2 方程 $\rightarrow$ 交点: $M_1(1, 2, \frac{5}{3}), M_2(-1, 2, \frac{5}{3})$

$M_1(1, 2, \frac{5}{3})$: $k_1 = \frac{\partial z}{\partial x} \Big|_{x=1} = 2x = 2 \quad k_2 = \frac{\partial z}{\partial x} \Big|_{x=1} = \frac{2}{3} \times 1 = \frac{2}{3}$

$$\theta_1 = \arctan \left| \frac{k_1 - k_2}{1 + k_1 k_2} \right| = \arctan \left| \frac{\frac{4}{3}}{1 + \frac{2}{3}} \right| = \arctan \frac{4}{7}$$

$M_2(-1, 2, \frac{5}{3}) \quad k_1 = -2 \quad k_2 = -\frac{2}{3}$

$$\theta_2 = \arctan \frac{4}{7}$$

综上, 两条平面曲线夹角为 $\arctan \frac{4}{7}$.

P₁₃ 习题8 P₃ 1, 3

I) $z = x \ln(xy)$

$$\frac{\partial z}{\partial x} = \ln(xy) + 1 \quad \frac{\partial^2 z}{\partial x^2} = \frac{1}{x} \quad \frac{\partial^2 z}{\partial x \partial y} = 0$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{1}{y} \quad \frac{\partial^2 z}{\partial x \partial y} = -\frac{1}{y^2}$$

$$\text{III). } u = x^3 + y^3 + z^3 - 3xyz$$

$$\frac{\partial u}{\partial x} = 3x^2 - 3yz \quad \frac{\partial u}{\partial y} = 3y^2 - 3xz \quad \frac{\partial u}{\partial z} = 3z^2 - 3xy$$

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 = 9[x^4 + y^4 + z^4 - 2xyz(x+y+z) + x^2y^2 + y^2z^2 + x^2z^2]$$

$$\frac{\partial^2 u}{\partial x^2} = 6x \quad \frac{\partial^2 u}{\partial y^2} = 6y \quad \frac{\partial^2 u}{\partial z^2} = 6z$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 6(x+y+z)$$

P₁₄ 习题 8 P₁₄ 3, 4

$$\text{III). } z = \ln(e^x + e^y)$$

$$\frac{\partial z}{\partial x} = \frac{e^x}{e^x + e^y} \quad \frac{\partial^2 z}{\partial x^2} = \frac{e^x(e^x + e^y) - e^{2x}}{(e^x + e^y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{e^y}{e^x + e^y} \quad \frac{\partial^2 z}{\partial y^2} = \frac{e^y(e^x + e^y) - e^{2y}}{(e^x + e^y)^2} \quad \rightarrow \quad \frac{\partial^2 z}{\partial x^2} \cdot \frac{\partial^2 z}{\partial y^2} - \left(\frac{\partial^2 z}{\partial x \partial y}\right)^2 = \frac{e^{2x}e^{2y}}{(e^x + e^y)^4} - \frac{e^{2x}e^{2y}}{(e^x + e^y)^4} = 0 \quad \square$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{-e^{x+y}}{(e^x + e^y)^2}$$

$$\text{IV). } r = \sqrt{x^2 + y^2 + z^2}$$

$$\ln r = \frac{1}{2} \ln(x^2 + y^2 + z^2)$$

$$\frac{\partial(\ln r)}{\partial x} = \frac{x}{x^2 + y^2 + z^2} = \frac{x}{r^2} \quad \frac{\partial^2(\ln r)}{\partial x^2} = \frac{r^2 - x \cdot 2x \cdot \frac{1}{r}}{r^4} = \frac{r^2 - 2x^2}{r^4}$$

$$\text{同理, } \frac{\partial^2(\ln r)}{\partial y^2} = \frac{r^2 - 2y^2}{r^4}, \quad \frac{\partial^2(\ln r)}{\partial z^2} = \frac{r^2 - 2z^2}{r^4}$$

$$\frac{\partial^2(\ln r)}{\partial x^2} + \frac{\partial^2(\ln r)}{\partial y^2} + \frac{\partial^2(\ln r)}{\partial z^2} = \frac{3r^2 - 2x^2 - 2y^2 - 2z^2}{r^4} = \frac{1}{r^2} \quad \square$$

P₁₅ 习题 8 P₁₆ 1, 4

$$\text{I). } z = \ln(x^2 + y^2)$$

$$\frac{\partial z}{\partial x} = \frac{2x}{x^2 + y^2} \quad \frac{\partial z}{\partial y} = \frac{2y}{x^2 + y^2}$$

$$dz = \frac{z(xdx + ydy)}{(x^2 + y^2)}$$

$$IV) u = \left(\frac{x}{y}\right)^{\frac{1}{z}}$$

$$\frac{\partial u}{\partial x} = \frac{1}{z} \cdot \frac{1}{y} \left(\frac{x}{y}\right)^{\frac{1}{z}-1} \quad \frac{\partial u}{\partial y} = \frac{1}{z} \cdot \left(-\frac{x}{y^2}\right) \left(\frac{x}{y}\right)^{\frac{1}{z}-1} \quad \frac{\partial u}{\partial z} = -\frac{1}{z^2} \ln \frac{x}{y} \cdot \left(\frac{x}{y}\right)^{\frac{1}{z}}$$

$$du = \frac{1}{yz} \left(\frac{x}{y}\right)^{\frac{1}{z}-1} dx - \frac{x}{y^{\frac{3}{2}}} \left(\frac{x}{y}\right)^{\frac{1}{z}-1} dy - \frac{1}{z^2} \ln \frac{x}{y} \left(\frac{x}{y}\right)^{\frac{1}{z}} dz$$

P16 题8 P17

$$I). \sqrt{1.02^2 + 1.97^2}$$

$$\text{令 } z = \sqrt{x^2 + y^2} \quad \text{取 } x_0 = 1, \quad \Delta x = 0.02, \quad y_0 = 2, \quad \Delta y = -0.03$$

$$\frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} \quad \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$$

$$z \approx f(1, 2) + f_x(1, 2) \Delta x + f_y(1, 2) \Delta y$$

$$= 2.218$$

$$II). 10.1^{2.03}$$

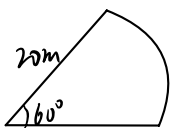
$$\text{令 } z = x^y \quad \text{取 } x_0 = 10, \quad \Delta x = 0.1, \quad y_0 = 2, \quad \Delta y = 0.03$$

$$\frac{\partial z}{\partial x} = y x^{y-1} \quad \frac{\partial z}{\partial y} = x^y \ln x$$

$$z \approx f(10, 2) + f_x(10, 2) \Delta x + f_y(10, 2) \Delta y$$

$$= 108.908$$

P17 题8 P19



$$S = \frac{\alpha}{360} \cdot \pi R^2 = \frac{\pi \alpha R^2}{360}$$

$$\text{有 } \alpha_0 = 60^\circ, \quad \Delta \alpha = 1^\circ, \quad R_0 = 20m$$

$$\frac{\partial S}{\partial \alpha} = \frac{\pi}{360} R^2 \quad \frac{\partial S}{\partial R} = \frac{\pi \alpha R}{180}$$

$$dS = \frac{\pi}{360} R^2 \Delta \alpha + \frac{\pi \alpha R}{180} \Delta R = 0$$

$$\text{解得 } \Delta R \approx -0.167m$$

P₁₉ 补充 8 P₃

$$f(x,y) = \begin{cases} (x^2+y^2) \sin \frac{1}{\sqrt{x^2+y^2}} & , x^2+y^2 \neq 0 \\ 0 & , x^2+y^2 = 0 \end{cases}$$

$$x^2+y^2 \neq 0: f_x(x,y) = 2x \sin \frac{1}{\sqrt{x^2+y^2}} - \frac{2x}{\sqrt{x^2+y^2}} \cos \frac{1}{\sqrt{x^2+y^2}}$$

$$f_y(x,y) = 2y \sin \frac{1}{\sqrt{x^2+y^2}} - \frac{2y}{\sqrt{x^2+y^2}} \cos \frac{1}{\sqrt{x^2+y^2}}$$

$$x^2+y^2 = 0 \quad f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(\Delta x, 0) - f(0,0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} 0 \Delta x \sin(\frac{1}{\Delta x})^2 = 0$$

$$\text{同理, } f_y(0,0) = 0$$

$$f_x(x,0) = 2x \sin \frac{1}{\sqrt{x}} - \frac{2x}{\sqrt{x}} \cos \frac{1}{\sqrt{x}} \quad x \rightarrow 0 \text{ 时 } \frac{2}{\sqrt{x}} \cos \frac{1}{\sqrt{x}} \text{ 振荡} \rightarrow f_x(x,y) \text{ 在 } (0,0) \text{ 不连续}$$

同理, $f_y(x,y)$ 在 $(0,0)$ 不连续

$$\lim_{\rho \rightarrow 0} \frac{\Delta f}{\rho} = \frac{(x^2+y^2) \sin \frac{1}{\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}} = \lim_{\rho \rightarrow 0} \rho \sin \frac{1}{\rho}$$

$$\rho \rightarrow 0 \text{ 时, } \rho \rightarrow 0, \text{ 而 } \sin \frac{1}{\rho} \text{ 有界} \rightarrow \lim_{\rho \rightarrow 0} \frac{\Delta f}{\rho} = 0, \Delta f = o(\rho)$$

故 $f(x,y)$ 在 $(0,0)$ 处可微 $\square$.

P₂₀ 补充 8 P₄

$$z(x,y) = \int_0^1 f(t) |xy-t| dt$$

$$= \int_0^{xy} f(t) (xy-t) dt + \int_{xy}^1 f(t) (t-xy) dt$$

$$\frac{\partial z}{\partial(xy)} = \int_0^{xy} f(t) dt - \int_{xy}^1 f(t) dt$$

$$\text{由于 } \frac{\partial xy}{\partial x} = y \xrightarrow{\text{链式}} \frac{\partial z}{\partial x} = y \left[\int_0^{xy} f(t) dt - \int_{xy}^1 f(t) dt \right]$$

$$\frac{\partial z}{\partial x} = y [f(xy)y + f(xy)y] = 2y^2 f(xy).$$

思考题

$$I). f(x,y) = \begin{cases} \frac{ye^{-\frac{x}{y^2}}}{e^{-\frac{x}{y^2}} + y^2} & x^2 + y^2 \neq 0 \\ 0 & x^2 + y^2 = 0 \end{cases}$$

两重极限: 令 $y=0 \rightarrow f(x,0)=0$

$$\text{令 } y=e^{-\frac{1}{x}} \rightarrow \frac{e^{-\frac{x}{e^{-\frac{1}{x}}}}}{e^{-\frac{x}{e^{-\frac{1}{x}}}} + e^{-\frac{2}{x}}} = \frac{1}{2} \rightarrow \text{二重极限不存在.}$$

累次极限: $\lim_{y \rightarrow 0} f(x,y) = 0$ $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x,y) = 0$ $\rightarrow$ 累次极限为 0

$$\lim_{x \rightarrow 0} f(x,y) \stackrel{z=\frac{x}{y^2}}{\rightarrow} \lim_{z \rightarrow \infty} \frac{ye^{-z}}{e^{-z} + y^2} = 0 \quad \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x,y) = 0$$

$$II). f(x,y) = \begin{cases} y + x \sin \frac{1}{y} & , y \neq 0 \\ 0 & , y = 0 \end{cases}$$

二重极限: $\sin \frac{1}{y}$ 有界 $|\sin \frac{1}{y}| \leq 1$

$$|y + x \sin \frac{1}{y}| \leq |y| + |x| \rightarrow \lim_{(x,y) \rightarrow (0,0)} y + x \sin \frac{1}{y} = 0$$

$(x,y) \rightarrow (0,0)$ 时, $|x| + |y| \rightarrow 0$

累次极限: 固定 $x \neq 0$, $y \rightarrow 0$ 时 $x \sin \frac{1}{y}$ 振荡 $\rightarrow \lim_{y \rightarrow 0} f(x,y)$ 不存在, $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x,y)$ 不存在

固定 $y \neq 0$, $x \rightarrow 0$ 时 $\sin \frac{1}{y}$ 为有量, x 是无穷量 $\rightarrow \lim_{x \rightarrow 0} f(x,y) = y$ $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x,y) = 0$.

$$III). f(x,y) = \begin{cases} \frac{xy}{x^2 + y^2} + x \sin \frac{1}{y} & , y \neq 0 \\ 0 & , y = 0 \end{cases}$$

两重极限: 令 $x=0 \rightarrow \frac{0 \cdot y}{0 + y^2} + 0 \cdot \sin \frac{1}{y} = 0$

$$\text{令 } y=kx \rightarrow f(x,y) = \frac{kx^2}{x^2 + k^2x^2} + x \sin \frac{1}{kx}$$

$\rightarrow$ 二重极限不存在.

$$\lim_{x \rightarrow 0} \frac{kx^2}{x^2 + k^2x^2} + x \sin \frac{1}{kx} = \frac{k}{1+k^2} \text{ 该值由 } k \text{ 决定}$$

累次极限 固定 $x \neq 0$ $y \rightarrow 0$ 时, $x \sin \frac{1}{y}$ 振荡 $\rightarrow \lim_{y \rightarrow 0} f(x,y)$ 不存在 $\rightarrow \lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x,y)$ 不存在

固定 $y \neq 0$ $x \rightarrow 0$ 时, $\sin \frac{1}{y}$ 为有量, x 是无穷量 $\rightarrow \lim_{x \rightarrow 0} f(x,y) = \frac{0 \cdot y}{0 + y^2} + 0 = 0 \rightarrow \lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x,y) = 0$

$$\text{IV). } f(x, y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2}, & x^2 + y^2 \neq 0 \\ 0, & x^2 + y^2 = 0 \end{cases}$$

二重极限: 令 $y=kx \rightarrow f(x, y) = \frac{x^2(1-k^2)}{x^2(1+k^2)} = \frac{1-k^2}{1+k^2}$, 极限与 k 有关 $\rightarrow$ 二重极限不存在

累次极限: 固定 $x \neq 0$, $y \rightarrow 0$ 时, $\lim_{y \rightarrow 0} f(x, y) = 1$, $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y) = 1$

固定 $y \neq 0$, $x \rightarrow 0$ 时, $\lim_{x \rightarrow 0} f(x, y) = -1$, $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y) = -1$.